

Pramo Jewels

UPI Payment Flow

Document ID: PJ-UPI-001

Version: 1.0

Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define the end-to-end UPI payment workflow and integration requirements
Audience	Backend Developers, Payment Team, QA, DevOps
Version	1.0

UPI Payment Lifecycle

Stage	Description
1. Checkout	Customer selects UPI as payment method.
2. Create Order	Backend creates payment order with gateway.
3. Collect Request	Customer approves payment in UPI app.
4. Gateway Callback	Gateway returns preliminary status.
5. Webhook Verification	Verify signature and payment authenticity.
6. Order Confirmation	Mark order as paid and reserve inventory.
7. Invoice	Generate invoice and notify customer.
8. Reconciliation	Daily settlement verification with gateway.

Supported Flows

Support UPI Intent, Collect Request and QR-based payments depending on gateway capabilities.

Business Rules

Create a unique payment reference, use idempotency keys, expire unpaid orders after a configurable timeout and prevent duplicate captures.

Security

Use HTTPS/TLS, validate webhook signatures, protect API credentials in a secrets manager, encrypt sensitive records and never store UPI PINs.

Failure Handling

Handle pending, failed, cancelled and timed-out transactions with automatic reconciliation and customer-friendly retry options.

Refund Flow

Initiate full or partial refunds through the gateway and synchronize refund status with the order lifecycle.

Monitoring

Track payment success rate, latency, webhook failures, pending transactions, reconciliation mismatches and refund turnaround time.

Integrations

Integrates with Cart, Checkout, Orders, Inventory, Notifications, Accounting, Analytics and Fraud Monitoring.

Acceptance Criteria

UPI payments are processed securely, payment states remain consistent, reconciliation succeeds and all events are audit logged.